

NITROGEN FAMILY - MASTER STATEMENT ASSIGNMENT

20 Questions | 15 Statements Each

Q1. OCCURRENCE, MEMBERS AND ELECTRONIC CONFIGURATION

How many of the following statements are correct?
_____ / 15

- (1) Group 15 consists of nitrogen, phosphorus, arsenic, antimony and bismuth.
- (2) Down the group there is a shift from non-metallic to metallic character through metalloidal character.
- (3) Nitrogen and phosphorus are non-metals; arsenic and antimony are metalloids; bismuth is a typical metal.
- (4) Molecular nitrogen constitutes about 78% by volume of the atmosphere.
- (5) In the earth's crust nitrogen occurs as NaNO_3 (Chile saltpetre) and KNO_3 (Indian saltpetre).
- (6) Nitrogen occurs in proteins of plants and animals.
- (7) Phosphorus occurs in apatite minerals $\text{Ca}_5(\text{PO}_4)_3\text{CaX}_2$, where $\text{X} = \text{F}, \text{Cl}$ or OH .
- (8) Fluorapatite is represented as $\text{Ca}_5(\text{PO}_4)_3\text{CaF}_2$.
- (9) Phosphorus is present in bones and living cells.
- (10) Phosphoproteins occur in milk and eggs.
- (11) Arsenic, antimony and bismuth are found mainly as sulphide minerals.
- (12) The general valence-shell configuration is $ns2np3$.
- (13) The s orbital is completely filled and p orbitals are half-filled, making the configuration extra stable.
- (14) The general valence-shell configuration is $ns2np5$.
- (15) Bismuth is classified as a metalloid in.

Q2. COMPLETE ATOMIC DATA

How many of the following statements are correct?
_____ / 15

- (1) Atomic numbers N, P, As, Sb, Bi are 7, 15, 33, 51, 83; atomic masses are 14.01, 30.97, 74.92, 121.75, 208.98 g mol⁻¹.
- (2) N has $[\text{He}]2s^2 2p^3$ and P has $[\text{Ne}]3s^2 3p^3$.
- (3) As has $[\text{Ar}]3d^{10} 4s^2 4p^3$.
- (4) Sb has $[\text{Kr}]4d^{10} 5s^2 5p^3$.
- (5) Bi has $[\text{Xe}]4f^{14} 5d^{10} 6s^2 6p^3$.
- (6) First ionisation enthalpies (kJ mol⁻¹) are 1402, 1012, 947, 834, 703 from N to Bi.
- (7) Second ionisation enthalpies (kJ mol⁻¹) are 2856, 1903, 1798, 1595, 1610.
- (8) Third ionisation enthalpies (kJ mol⁻¹) are 4577, 2910, 2736, 2443, 2466.
- (9) Electronegativities are 3.0, 2.1, 2.0, 1.9, 1.9 from N to Bi.
- (10) Covalent radii (pm) are 70, 110, 121, 141, 148.
- (11) The ionic-radius entries are N^{3-} 171 pm, P^{3-} 212 pm, As^{3-} 222 pm, Sb^{3+} 76 pm and Bi^{3+} 103 pm.
- (12) The first ionisation enthalpy of Bi is larger than that of N.
- (13) The covalent radius of P is less than that of N.
- (14) Sb and Bi have the same tabulated electronegativity, 1.9.
- (15) N has the smallest tabulated covalent radius in the group.

Q3. MELTING POINT, BOILING POINT, DENSITY AND PERIODIC TRENDS

How many of the following statements are correct?
_____ / 15

- (1) Melting points (K) given for N_2 , white P, grey α -As, Sb and Bi are about 63, 317, 1089, 904 and 544.
- (2) Boiling points (K) given are about 77.2, 554, 888, 1860 and 1837; the As value is a sublimation temperature.

- (3) Densities (g cm⁻³) given are 0.879 for N_2 at 63 K, 1.823 for P, 5.778 for grey α -As, 6.697 for Sb and 9.808 for Bi.
- (4) Covalent and ionic radii generally increase down Group 15.
- (5) There is a considerable increase in covalent radius from N to P.
- (6) From As to Bi only a small increase in covalent radius is observed because of filled d and/or f orbitals.
- (7) Ionisation enthalpy generally decreases down the group because atomic size increases.
- (8) Group 15 has higher ionisation enthalpy than corresponding Group 14 elements because of smaller size and extra-stable half-filled p orbitals.
- (9) For a given element, $\Delta_i H_1 < \Delta_i H_2 < \Delta_i H_3$.
- (10) Electronegativity generally decreases down the group; differences among heavier members are small.
- (11) Dinitrogen is a diatomic gas, whereas the other members are solids.
- (12) Metallic character increases down the group.
- (13) Boiling points generally increase down the group.
- (14) Melting point increases up to arsenic and then decreases up to bismuth.
- (15) Except nitrogen, all Group 15 elements show allotropy.

Q4. OXIDATION STATES, INERT-PAIR EFFECT AND DISPROPORTIONATION

How many of the following statements are correct?
_____ / 15

- (1) Common oxidation states are -3, +3 and +5.
- (2) The tendency to show -3 oxidation state decreases down the group.
- (3) Bismuth hardly forms any compound in the -3 oxidation state.
- (4) Stability of +5 oxidation state decreases down the group.
- (5) Stability of +3 oxidation state increases down the group due to inert-pair effect.
- (6) BiF_5 is cited as the only well-characterised Bi(V) compound.
- (7) Nitrogen also shows +1, +2 and +4 oxidation states with oxygen.
- (8) Phosphorus also shows +1 and +4 oxidation states in some oxoacids.
- (9) Nitrogen oxidation states from +1 to +4 tend to disproportionate in acid solution.
- (10) $3\text{HNO}_2 \rightarrow \text{HNO}_3 + \text{H}_2\text{O} + 2\text{NO}$ is an disproportionation example.
- (11) Nearly all intermediate oxidation states of phosphorus disproportionate to +5 and -3 in acid and alkali.
- (12) +3 oxidation state of As, Sb and Bi becomes increasingly stable toward disproportionation.
- (13) Stability of +5 increases down the group.
- (14) Bismuth readily forms many stable -3 compounds.
- (15) Phosphorus is restricted only to -3, +3 and +5.

Q5. ANOMALOUS NITROGEN, COVALENCY, MULTIPLE BONDING AND CATENATION

How many of the following statements are correct?
_____ / 15

- (1) Nitrogen has maximum covalency 4 because only one s and three p orbitals are available for bonding.
- (2) Heavier members can expand covalency using vacant d orbitals in the explanation; PF_6^- is cited.
- (3) Nitrogen differs because of small size, high electronegativity, high ionisation enthalpy and absence of d orbitals.
- (4) Nitrogen forms π - π multiple bonds with itself and with small electronegative atoms such as C and O.

- (5) Heavier members do not form effective $p\pi-p\pi$ bonds because their orbitals are large and diffuse.
- (6) N_2 has a triple bond consisting of one sigma and two pi bonds.
- (7) The $N \equiv N$ bond enthalpy is about 941.4 kJ mol⁻¹.
- (8) P, As and Sb form single P-P, As-As and Sb-Sb bonds in elemental states, while Bi forms metallic bonds.
- (9) A single N-N bond is weaker than a single P-P bond because non-bonding-electron repulsion is high at short N-N distance.
- (10) Therefore nitrogen has weaker catenation tendency than phosphorus.
- (11) Nitrogen cannot form $d\pi-p\pi$ bonds like $R_3P=O$ or $R_3P=CH_2$.
- (12) P and As can form $d\pi-d\pi$ bonds with transition metals when $P(C_2H_5)_3$ and $As(C_6H_5)_3$ act as ligands.
- (13) Nitrogen has stronger catenation than phosphorus.
- (14) Nitrogen forms stable pentahalides by using vacant d orbitals.
- (15) $R_3P=O$ can exist whereas the analogous $R_3N=O$ is not explained by d-orbital participation for nitrogen.

Q6. HYDRIDES - ALL TABULATED VALUES AND TRENDS

How many of the following statements are correct?
_____ / 15

- (1) All Group 15 elements form EH_3 hydrides.
- (2) Hydride stability decreases $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$.
- (3) Reducing character increases down the group; NH_3 is mild and BiH_3 is strongest among these hydrides.
- (4) Basicity decreases $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$.
- (5) Melting points (K) of NH_3 , PH_3 , AsH_3 and SbH_3 are 195.2, 139.5, 156.7 and 185.
- (6) Boiling points (K) of NH_3 , PH_3 , AsH_3 , SbH_3 , BiH_3 are 238.5, 185.5, 210.6, 254.6, 290.
- (7) E-H distances (pm) of NH_3 , PH_3 , AsH_3 and SbH_3 are 101.7, 141.9, 151.9 and 170.7.
- (8) H-E-H angles are 107.8°, 93.6°, 91.8° and 91.3° for NH_3 , PH_3 , AsH_3 and SbH_3 .
- (9) $\Delta_f H^\circ$ values (kJ mol⁻¹) from NH_3 to BiH_3 are -46.1, 13.4, 66.4, 145.1 and 278.
- (10) E-H bond dissociation enthalpies (kJ mol⁻¹) for NH_3 , PH_3 , AsH_3 , SbH_3 are 389, 322, 297 and 255.
- (11) PH_3 has lower boiling point than NH_3 because PH_3 lacks intermolecular hydrogen bonding.
- (12) The HNH angle is larger than HPH, HAsH and HSbH.
- (13) BiH_3 is the least stable hydride.
- (14) Basicity increases from NH_3 to BiH_3 .
- (15) HPH angle is larger than HNH angle.

Q7. REACTIVITY TOWARD OXYGEN, HALOGENS AND METALS

How many of the following statements are correct?
_____ / 15

- (1) Group 15 elements form E_2O_3 and E_2O_5 oxides.
- (2) The oxide in the higher oxidation state is more acidic than the lower oxidation-state oxide.
- (3) Acidic character of oxides decreases down the group.
- (4) E_2O_3 of N and P are purely acidic.
- (5) E_2O_3 of As and Sb are amphoteric.
- (6) The corresponding Bi oxide is predominantly basic.
- (7) Group 15 forms EX_3 and EX_5 halides.
- (8) Nitrogen does not form pentahalides because it lacks d orbitals in its valence shell.
- (9) Pentahalides are more covalent than trihalides.
- (10) All trihalides except those of nitrogen are stable; for nitrogen only NF_3 is stable.

- (11) Trihalides except BiF_3 are predominantly covalent.
- (12) Reaction with metals gives binary compounds in which the Group 15 element has -3 oxidation state.
- (13) Examples include Ca_3N_2 , Ca_3P_2 , sodium arsenide, Zn_3Sb_2 and Mg_3Bi_2 .
- (14) Acidic character of Group 15 oxides increases down the group.
- (15) NCl_3 is the only stable nitrogen trihalide according to the chapter.

Q8. DINITROGEN - EVERY METHOD OF PREPARATION

How many of the following statements are correct?
_____ / 15

- (1) Commercial N_2 is obtained by liquefaction and fractional distillation of air.
- (2) Liquid N_2 , b.p. 77.2 K, distils before liquid O_2 , b.p. about 90 K.
- (3) Laboratory N_2 is prepared by treating aqueous NH_4Cl with $NaNO_2$.
- (4) $NH_4Cl + NaNO_2 \rightarrow N_2 + 2H_2O + NaCl$.
- (5) Small amounts of NO and HNO_3 are also formed in this laboratory preparation.
- (6) The impurities are removed by passing the gas through aqueous H_2SO_4 containing $K_2Cr_2O_7$.
- (7) Thermal decomposition of ammonium dichromate also gives N_2 .
- (8) $(NH_4)_2Cr_2O_7 \rightarrow N_2 + 4H_2O + Cr_2O_3$ on heating.
- (9) Very pure N_2 can be obtained by thermal decomposition of sodium azide.
- (10) Very pure N_2 can also be obtained from barium azide.
- (11) $2NaN_3 \rightarrow 2Na + 3N_2$.
- (12) $Ba(N_3)_2 \rightarrow Ba + 3N_2$.
- (13) During fractional distillation O_2 distils before N_2 .
- (14) $K_2Cr_2O_7/H_2SO_4$ is used to generate N_2 rather than purify it.
- (15) The azide method is specifically described as giving very pure nitrogen.

Q9. DINITROGEN - PROPERTIES, REACTIONS AND USES

How many of the following statements are correct?
_____ / 15

- (1) N_2 is colourless, odourless, tasteless and non-toxic.
- (2) Nitrogen has two stable isotopes, ^{14}N and ^{15}N .
- (3) Solubility of N_2 is about 23.2 cm³ per litre of water at 273 K and 1 bar.
- (4) N_2 has low freezing and boiling points.
- (5) It is rather inert at room temperature because of the very high $N \equiv N$ bond enthalpy.
- (6) Reactivity increases rapidly with rise in temperature.
- (7) At high temperature, some metals form predominantly ionic nitrides.
- (8) At high temperature, non-metals form covalent nitrides.
- (9) $6Li + N_2 \rightarrow 2Li_3N$ on heating.
- (10) $3Mg + N_2 \rightarrow Mg_3N_2$ on heating.
- (11) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ at about 773 K in presence of catalyst.
- (12) $N_2 + O_2 \rightarrow 2NO$ only at very high temperature, about 2000 K.
- (13) N_2 is used to manufacture NH_3 and nitrogen compounds such as calcium cyanamide.
- (14) N_2 is used as an inert atmosphere; liquid N_2 is used for refrigeration, preservation and cryosurgery.
- (15) N_2 is highly reactive at room temperature because its triple bond is weak.

Q10. AMMONIA - OCCURRENCE, PREPARATION AND HABER PROCESS

How many of the following statements are correct?
_____ / 15

- (1) NH_3 is present in small quantities in air and soil and forms from decay of nitrogenous organic matter such as urea.
- (2) On a small scale NH_3 is obtained by treating ammonium salts with NaOH or Ca(OH)_2 .
- (3) $2\text{NH}_4\text{Cl} + \text{Ca(OH)}_2 \rightarrow 2\text{NH}_3 + 2\text{H}_2\text{O} + \text{CaCl}_2$.
- (4) $(\text{NH}_4)_2\text{SO}_4 + 2\text{NaOH} \rightarrow 2\text{NH}_3 + 2\text{H}_2\text{O} + \text{Na}_2\text{SO}_4$.
- (5) Industrial NH_3 is manufactured by Haber process.
- (6) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$.
- (7) The enthalpy value shown is $\Delta H^\circ = -46.1 \text{ kJ mol}^{-1}$.
- (8) High pressure favours NH_3 formation according to Le Chatelier's principle.
- (9) Optimum pressure quoted is $200 \times 10^5 \text{ Pa}$, about 200 atm.
- (10) Optimum temperature quoted is about 700 K.
- (11) Iron oxide with small amounts of K_2O and Al_2O_3 is mentioned for faster attainment of equilibrium.
- (12) Earlier iron was used as catalyst with molybdenum as promoter.
- (13) 's flow chart operates around 20 MPa and 700 K in the synthesis loop.
- (14) Low pressure favours NH_3 because more gas moles are formed.
- (15) The optimum temperature quoted is about 2000 K.

Q11. AMMONIA - PROPERTIES, COMPLEXES, REACTIONS AND ALL USES

How many of the following statements are correct?
_____ / 15

- (1) NH_3 is a colourless gas with pungent odour; f.p. 198.4 K and b.p. 239.7 K.
- (2) Solid and liquid NH_3 are hydrogen-bonded, giving higher melting/boiling points than expected from molecular mass.
- (3) NH_3 is trigonal pyramidal with three bond pairs and one lone pair on N.
- (4) NH_3 is highly soluble in water and its solution is weakly basic: $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$.
- (5) NH_3 forms ammonium salts with acids.
- (6) As a weak base it precipitates hydroxides/hydrated oxides of many metal ions.
- (7) The N lone pair makes NH_3 a Lewis base; this is the basis of.
- (8) $\text{Cu}^{2+} + 4\text{NH}_3$ gives deep-blue $[\text{Cu}(\text{NH}_3)_4]^{2+}$.
- (9) White AgCl dissolves in NH_3 to give a colourless $[\text{Ag}(\text{NH}_3)_2]^+$ complex.
- (10) Such complex formation is used in detection of Cu^{2+} and Ag^+ .
- (11) NH_3 is used to produce ammonium nitrate, urea, ammonium phosphate and ammonium sulphate.
- (12) NH_3 is used to prepare inorganic nitrogen compounds, especially HNO_3 .
- (13) Liquid NH_3 is used as a refrigerant.
- (14) NH_3 is planar and contains no lone pair.
- (15) The Ag-NH_3 complex is deep blue.

Q12. OXIDES OF NITROGEN: PREPARATION, COLOUR AND NATURE

How many of the following statements are correct?
_____ / 15

- (1) N_2O is dinitrogen oxide/nitrogen(I) oxide; it is a colourless neutral gas.
- (2) N_2O is prepared by heating NH_4NO_3 .
- (3) NO is nitrogen monoxide/nitrogen(II) oxide; it is a colourless neutral gas.

(4) NO is prepared in the from NaNO_2 , FeSO_4 and H_2SO_4 .

(5) N_2O_3 is dinitrogen trioxide/nitrogen(III) oxide; it is a blue acidic solid.

(6) N_2O_3 is formed at about 250 K from NO and N_2O_4 .

(7) NO_2 is nitrogen dioxide/nitrogen(IV) oxide; it is a brown acidic gas.

(8) NO_2 is prepared by heating $\text{Pb(NO}_3)_2$ at about 673 K.

(9) N_2O_4 is dinitrogen tetroxide/nitrogen(IV) oxide; it is a colourless acidic solid/liquid.

(10) $2\text{NO}_2 \rightleftharpoons \text{N}_2\text{O}_4$; cooling favours N_2O_4 and heating favours NO_2 .

(11) N_2O_5 is dinitrogen pentoxide/nitrogen(V) oxide; it is a colourless acidic solid.

(12) N_2O_5 can be prepared using HNO_3 and P_4O_{10} .

(13) NO_2 has an odd number of valence electrons and dimerises to even-electron N_2O_4 .

(14) uses this odd-electron argument to explain dimerisation.

(15) N_2O_4 is always a brown neutral gas.

Q13. RESONANCE STRUCTURES, BOND LENGTHS AND BOND ANGLES

How many of the following statements are correct?
_____ / 15

(1) N_2O is linear with $\text{N-N} \approx 113 \text{ pm}$ and $\text{N-O} \approx 119 \text{ pm}$.

(2) NO has N-O bond length $\approx 115 \text{ pm}$.

(3) N_2O_3 is planar.

(4) For N_2O_3 the shows $\text{N-N} \approx 186 \text{ pm}$, one N-O value $\approx 114 \text{ pm}$ and another $\approx 121 \text{ pm}$.

(5) Angles shown for N_2O_3 include about 105° , 130° and 117° .

(6) NO_2 is angular with $\text{N-O} \approx 120 \text{ pm}$ and $\text{O-N-O} \approx 134^\circ$.

(7) N_2O_4 is planar with $\text{N-N} \approx 175 \text{ pm}$.

(8) N_2O_4 shows $\text{N-O} \approx 121 \text{ pm}$ and an O-N-O angle $\approx 135^\circ$.

(9) N_2O_5 is planar in the molecular structure shown.

(10) In N_2O_5 the bridging N-O bond is $\approx 151 \text{ pm}$ and terminal N-O is $\approx 119 \text{ pm}$.

(11) N_2O_5 shows an N-O-N angle $\approx 112^\circ$ and an O-N-O angle $\approx 134^\circ$.

(12) displays main Lewis resonance structures for the oxides.

(13) Covalence of nitrogen in N_2O_5 is four, as is.

(14) NO_2 is linear with bond angle 180° .

(15) The N-O bond in NO is longer than 200 pm.

Q14. NITRIC ACID - OXOACIDS, LABORATORY PREPARATION AND OSTWALD PROCESS

How many of the following statements are correct?
_____ / 15

(1) Nitrogen oxoacids given include $\text{H}_2\text{N}_2\text{O}_2$, HNO_2 and HNO_3 ; HNO_3 is the most important.

(2) Laboratory HNO_3 is prepared by heating KNO_3 or NaNO_3 with concentrated H_2SO_4 in a glass retort.

(3) $\text{NaNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HNO}_3$.

(4) Large-scale HNO_3 is prepared mainly by Ostwald process.

(5) Ostwald process is based on catalytic oxidation of NH_3 by atmospheric O_2 .

(6) Pt/Rh gauze is the catalyst stated for the first step.

(7) The stated first-step conditions are about 500 K and 9 bar.

(8) $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$.

(9) $2\text{NO} + \text{O}_2 \rightleftharpoons 2\text{NO}_2$.

(10) $3\text{NO}_2 + \text{H}_2\text{O} \rightarrow 2\text{HNO}_3 + \text{NO}$.

(11) NO formed in the absorption step is recycled.

(12) Aqueous HNO_3 is concentrated by distillation to about 68% by mass.

(13) Further concentration to about 98% is achieved by dehydration with concentrated H_2SO_4 .

- (14) Ostwald process begins by reducing nitrate to NH_3 .
 (15) The first catalytic oxidation is carried out over iron at about 2000 K.

Q15. NITRIC ACID - PROPERTIES, OXIDATION REACTIONS, BROWN-RING TEST AND USES

How many of the following statements are correct?
 _____ / 15

- (1) HNO_3 is a colourless liquid with f.p. 231.4 K and b.p. 355.6 K; laboratory-grade acid is about 68% by mass with specific gravity 1.504.
 (2) Gaseous HNO_3 is planar; in water it behaves as a strong acid: $\text{HNO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{NO}_3^-$.
 (3) Concentrated HNO_3 is a strong oxidising agent; it attacks most metals except noble metals such as Au and Pt, and products depend on concentration, temperature and nature of substrate.
 (4) $3\text{Cu} + 8\text{HNO}_3(\text{dil.}) \rightarrow 3\text{Cu}(\text{NO}_3)_2 + 2\text{NO} + 4\text{H}_2\text{O}$, whereas $\text{Cu} + 4\text{HNO}_3(\text{conc.}) \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O}$.
 (5) Dilute HNO_3 with Zn gives N_2O , whereas concentrated HNO_3 gives NO_2 ; Cr and Al may resist concentrated HNO_3 because of a passive oxide film.
 (6) Concentrated HNO_3 oxidises I_2 to HIO_3 , C to CO_2 , S to H_2SO_4 and P to H_3PO_4 .
 (7) gives $\text{I}_2 + 10\text{HNO}_3 \rightarrow 2\text{HIO}_3 + 10\text{NO}_2 + 4\text{H}_2\text{O}$ and $\text{C} + 4\text{HNO}_3 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + 4\text{NO}_2$.
 (8) gives $\text{S}_8 + 48\text{HNO}_3 \rightarrow 8\text{H}_2\text{SO}_4 + 48\text{NO}_2 + 16\text{H}_2\text{O}$ and $\text{P}_4 + 20\text{HNO}_3 \rightarrow 4\text{H}_3\text{PO}_4 + 20\text{NO}_2 + 4\text{H}_2\text{O}$.
 (9) Brown-ring test depends on Fe^{2+} reducing nitrate to NO, followed by formation of a brown nitrosyl complex with Fe^{2+} .
 (10) The test uses dilute FeSO_4 followed by careful addition of concentrated H_2SO_4 along the test-tube wall; a brown ring at the interface indicates nitrate.
 (11) $\text{NO}_3^- + 3\text{Fe}^{2+} + 4\text{H}^+ \rightarrow \text{NO} + 3\text{Fe}^{3+} + 2\text{H}_2\text{O}$; the brown complex is $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$.
 (12) Major HNO_3 uses include ammonium nitrate manufacture, nitrates for explosives/pyrotechnics, and preparation of nitroglycerin, TNT and other organic nitro compounds.
 (13) Other given uses are pickling stainless steel, etching metals and use as an oxidiser in rocket fuels.
 (14) asks why Cu gives different products with HNO_3 and asks for resonance structures of NO_2 and N_2O_5 .
 (15) The brown-ring test is based on direct reduction of nitrate to NH_3 , and the brown species is $[\text{Cu}(\text{NH}_3)_4]^{2+}$.

Q16. PHOSPHORUS - WHITE, RED, ALPHA-BLACK AND BETA-BLACK ALLOTROPES

How many of the following statements are correct?
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- (1) Important allotropes discussed are white, red and black phosphorus.
 (2) White P is a translucent white waxy poisonous solid, insoluble in water but soluble in CS_2 .
 (3) White P glows in dark by chemiluminescence.
 (4) Boiling white P with NaOH in inert atmosphere gives PH_3 and NaH_2PO_2 .
 (5) White P is less stable and more reactive because of angular strain in P_4 with 60° angles.
 (6) White P catches fire in air to give dense white fumes of P_4O_{10} .
 (7) White P consists of discrete tetrahedral P_4 molecules.
 (8) Red P is obtained by heating white P at 573 K in inert atmosphere for several days.
 (9) Red P has iron-grey lustre, is odourless, non-poisonous and insoluble in water and CS_2 .
 (10) Red P is much less reactive than white P and does not glow in dark.

- (11) Red P is polymeric, represented as chains of linked P_4 tetrahedra.
 (12) α -Black P is formed by heating red P in a sealed tube at 803 K.
 (13) α -Black P has opaque monoclinic/rhombohedral crystals and does not oxidise in air.
 (14) β -Black P is prepared by heating white P at 473 K under high pressure and does not burn in air up to 673 K.
 (15) White P is polymeric whereas red P consists of isolated P_4 molecules.

Q17. PHOSPHINE - EVERY PREPARATION, PURIFICATION, PROPERTY AND USE

How many of the following statements are correct?
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- (1) PH_3 is prepared by $\text{Ca}_3\text{P}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Ca}(\text{OH})_2 + 2\text{PH}_3$.
 (2) PH_3 is also prepared by $\text{Ca}_3\text{P}_2 + 6\text{HCl} \rightarrow 3\text{CaCl}_2 + 2\text{PH}_3$.
 (3) Laboratory PH_3 is prepared by heating white P with concentrated NaOH in inert CO_2 atmosphere.
 (4) This laboratory reaction also gives sodium hypophosphite, NaH_2PO_2 .
 (5) Pure PH_3 is non-inflammable but becomes inflammable due to P_2H_4 or P_4 vapours.
 (6) PH_3 is purified by absorption in HI to form PH_4I ; KOH then liberates PH_3 .
 (7) PH_3 is a colourless gas with rotten-fish smell and is highly poisonous.
 (8) PH_3 explodes with traces of oxidising agents such as HNO_3 , Cl_2 and Br_2 vapours.
 (9) PH_3 is slightly soluble in water.
 (10) Its aqueous solution decomposes in light giving red phosphorus and H_2 .
 (11) With CuSO_4 or HgCl_2 it gives corresponding phosphides.
 (12) PH_3 is weakly basic and forms phosphonium compounds such as PH_4Br with acids.
 (13) establishes the basic nature of PH_3 from formation of PH_4I and its lone pair.
 (14) Spontaneous combustion of PH_3 is used in Holme's signals; PH_3 is also used in smoke screens.
 (15) PH_3 is more strongly basic and more water-soluble than NH_3 .

Q18. PHOSPHORUS HALIDES - PREPARATION, HYDROLYSIS, STRUCTURES, BOND LENGTHS AND REACTIONS

How many of the following statements are correct?
 _____ / 15

- (1) Phosphorus forms PX_3 ($\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$) and PX_5 ($\text{X} = \text{F}, \text{Cl}, \text{Br}$).
 (2) PCl_3 is prepared by passing dry chlorine over heated white phosphorus: $\text{P}_4 + 6\text{Cl}_2 \rightarrow 4\text{PCl}_3$.
 (3) PCl_3 is also prepared by the action of thionyl chloride on white phosphorus: $\text{P}_4 + 8\text{SOCl}_2 \rightarrow 4\text{PCl}_3 + 4\text{SO}_2 + 2\text{S}_2\text{Cl}_2$.
 (4) PCl_3 is a colourless oily liquid and hydrolyses in moisture: $\text{PCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_3 + 3\text{HCl}$.
 (5) PCl_3 reacts with CH_3COOH to give CH_3COCl and with $\text{C}_2\text{H}_5\text{OH}$ to give $\text{C}_2\text{H}_5\text{Cl}$.
 (6) PCl_3 is pyramidal and phosphorus is described as sp^3 hybridised.
 (7) PCl_5 is prepared by treating white phosphorus with excess dry chlorine: $\text{P}_4 + 10\text{Cl}_2 \rightarrow 4\text{PCl}_5$.
 (8) PCl_5 is also prepared by the action of SO_2Cl_2 on phosphorus: $\text{P}_4 + 10\text{SO}_2\text{Cl}_2 \rightarrow 4\text{PCl}_5 + 10\text{SO}_2$.
 (9) PCl_5 is a yellowish-white powder; in moist air it first gives POCl_3 and finally H_3PO_4 .
 (10) On stronger heating, PCl_5 decomposes: $\text{PCl}_5 \rightarrow \text{PCl}_3 + \text{Cl}_2$.

(11) PCl_5 converts compounds containing $-\text{OH}$ groups into chloro derivatives and reacts with finely divided metals to give metal chlorides.

(12) In gaseous and liquid states PCl_5 is trigonal bipyramidal; three equatorial $\text{P}-\text{Cl}$ bonds are equivalent.

(13) Equatorial $\text{P}-\text{Cl}$ bonds are about 202 pm, while axial $\text{P}-\text{Cl}$ bonds are about 240 pm and are longer because axial bond pairs suffer greater repulsion.

(14) In solid state PCl_5 exists as $[\text{PCl}_4]^+[\text{PCl}_6]^-$; $[\text{PCl}_4]^+$ is tetrahedral and $[\text{PCl}_6]^-$ is octahedral.

(15) All five $\text{P}-\text{Cl}$ bonds in gaseous PCl_5 are equivalent and have the same bond length.

Q19. PHOSPHORUS OXOACIDS - FORMULAS, STRUCTURES, PREPARATION, BASICITY AND REDUCING BEHAVIOUR

How many of the following statements are correct?
_____ / 15

(1) H_3PO_2 has P in +1 oxidation state and contains one $\text{P}-\text{OH}$, two $\text{P}-\text{H}$ and one $\text{P}=\text{O}$ bond.

(2) H_3PO_3 has P in +3 oxidation state and contains two $\text{P}-\text{OH}$, one $\text{P}-\text{H}$ and one $\text{P}=\text{O}$ bond.

(3) $\text{H}_4\text{P}_2\text{O}_5$ has P in +3 oxidation state and contains two $\text{P}-\text{OH}$, two $\text{P}-\text{H}$ and two $\text{P}=\text{O}$ bonds.

(4) $\text{H}_4\text{P}_2\text{O}_6$ has P in +4 oxidation state and contains four $\text{P}-\text{OH}$, two $\text{P}=\text{O}$ and one $\text{P}-\text{P}$ bond.

(5) H_3PO_4 has P in +5 oxidation state and contains three $\text{P}-\text{OH}$ and one $\text{P}=\text{O}$ bond.

(6) $\text{H}_4\text{P}_2\text{O}_7$ has P in +5 oxidation state and contains four $\text{P}-\text{OH}$, two $\text{P}=\text{O}$ and one $\text{P}-\text{O}-\text{P}$ bond.

(7) Metaphosphoric acid is polymeric, $(\text{HPO}_3)_n$; the cyclic trimer contains $\text{P}-\text{OH}$, $\text{P}=\text{O}$ and $\text{P}-\text{O}-\text{P}$ linkages.

(8) H_3PO_2 is obtained from white phosphorus and alkali; H_3PO_3 is obtained from P_2O_3 and water.

(9) $\text{H}_4\text{P}_2\text{O}_5$ is prepared from PCl_3 and H_3PO_3 ; $\text{H}_4\text{P}_2\text{O}_6$ is obtained from red phosphorus and alkali.

(10) H_3PO_4 is obtained from P_4O_{10} and water; $\text{H}_4\text{P}_2\text{O}_7$ is obtained by heating phosphoric acid.

(11) In these oxoacids phosphorus is tetrahedrally surrounded by other atoms and every acid contains at least one $\text{P}=\text{O}$ and one $\text{P}-\text{OH}$ bond.

(12) For phosphorus oxidation states below +5, the oxoacid contains either a $\text{P}-\text{H}$ bond or a $\text{P}-\text{P}$ bond in addition to $\text{P}=\text{O}/\text{P}-\text{OH}$, but not both.

(13) On heating, $4\text{H}_3\text{PO}_3 \rightarrow 3\text{H}_3\text{PO}_4 + \text{PH}_3$; acids containing $\text{P}-\text{H}$ bonds show reducing character.

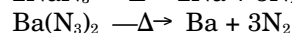
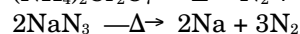
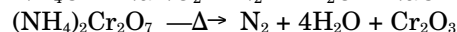
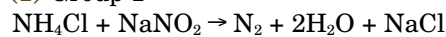
(14) H_3PO_2 reduces AgNO_3 to silver; $\text{P}-\text{H}$ hydrogens are not ionisable, so H_3PO_3 is dibasic and H_3PO_4 is tribasic.

(15) H_3PO_2 is tribasic because all three hydrogen atoms are ionisable.

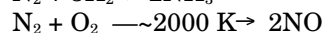
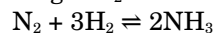
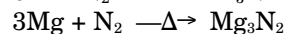
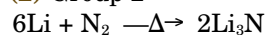
Q20. MASTER REACTION MATCHING - HOW MANY REACTION GROUPS ARE COMPLETELY CORRECT?

How many of the following statements are correct?
_____ / 15

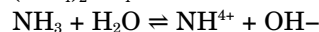
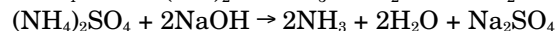
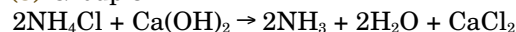
(1) Group 1



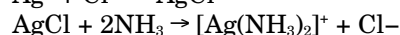
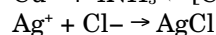
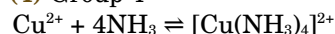
(2) Group 2



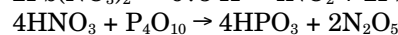
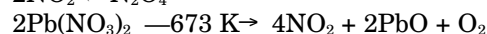
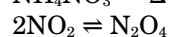
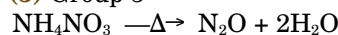
(3) Group 3



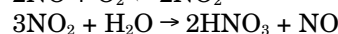
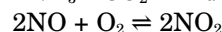
(4) Group 4



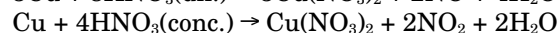
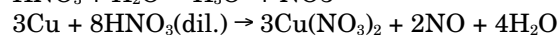
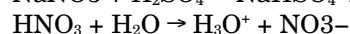
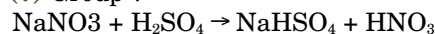
(5) Group 5



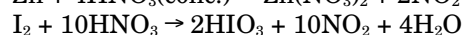
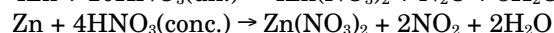
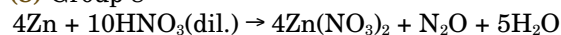
(6) Group 6



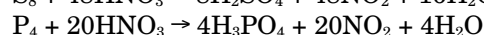
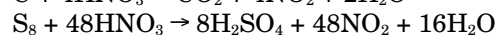
(7) Group 7



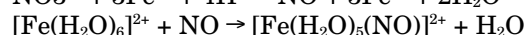
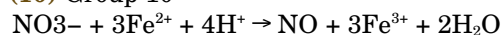
(8) Group 8



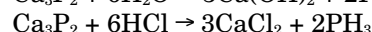
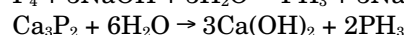
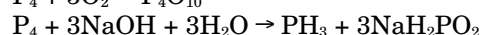
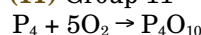
(9) Group 9



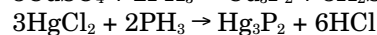
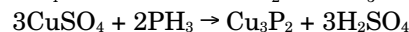
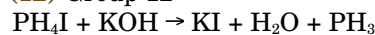
(10) Group 10



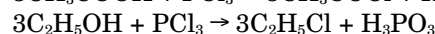
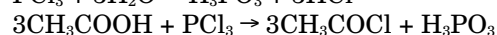
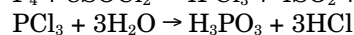
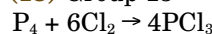
(11) Group 11



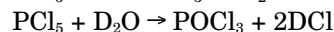
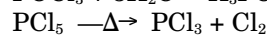
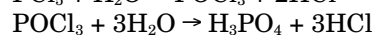
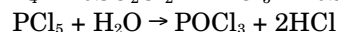
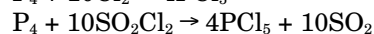
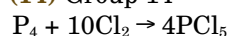
(12) Group 12



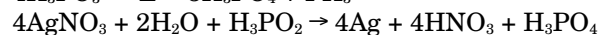
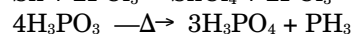
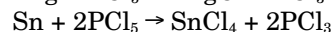
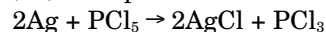
(13) Group 13



(14) Group 14



(15) Group 15



ANSWER KEY

Correct	Q
13	11
13	12
15	13
12	14
13	15
13	16
13	17
13	18
14	19
13	20